

AAMLEEN AHMED

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EDUCATION

Indraprastha Institute of Information Technology, Delhi (IIITD)

Aug, 2020 – June, 2024

B.Tech (Honors) - Computer Science and Engineering

CGPA - 8.86/10

New Delhi, India

COURSEWORK / SKILLS

- Distributed Systems
- Network Security
- Artificial Intelligence
- Natural Language Processing
- Cloud & DevOps
- Machine Learning
- Data Science & Analysis
- Data-Structures & Algorithms

TECHNICAL SKILLS

Languages: C++, C, Python, JAVA, JavaScript, SQL

Technology/Frameworks: Kubernetes, Hadoop, Kafka, SpringBoot, REST, Keras-Tensor, PyTorch, LLMs

Developer Tools: Google Cloud Platform, Terraform, Mockito, Jenkins, GitLabs, Linux, React, Hugging Face

EXPERIENCE

Software Development Engineer -1 at *Brane Enterprises*

Jul, 24 - Present

- Contributed to full development life-cycle in **Translator Microservice** for an AI-based platform, facilitating seamless translation between 35 languages to support development in natural spoken language.
- Improved system performance by designing API handlers to resolve issues & add features using Spring.
- Increased SonarQube code quality from 90% to 99%, using GitLab, Jenkins, Kafka, and Grafana.

Researcher at *Infosys-CAI & MAX-Healthcare* | Machine Learning

Aug, 23 - Jun, 24

- Innovated an **ML-DL Architecture** to detect breast cancer sub-types using MRI Images with 100+ features.
- Proposed a complex pipeline with supervised ML algorithms using Data Science & Ensemble Boosting, improving cancer detection scores to 80% from 55% baseline. Deployed with MAX in real world
- Project received **100% fellowship** from ARTPARK -IISc Bangalore. Submitted to MLHC Conference 2024.

Undergrad Researcher at *HiPEC Labs* | Parallel Processing

Aug, 22 - May, 23

- Researched & optimized a multi-thread work-stealing algorithm in **Linux kernel**, enhancing task distribution & minimizing idle threads. Tested on servers with 120+ threads & achieved **10% improvement**.
- Applied TSO buffers, enhancing **memory management** & synchronization by **15%** in parallel runtimes.

PROJECTS

RAFT, & Cloud Applications | Distributed Systems, DevOps, Networks

Feb,24 - Mar,24

- Implemented Cloud consensus RAFT algorithm from scratch. Managed leader election, logs & interactions.
- Created distributed applications: Ecom Market (gRPC), Group Messaging System (ZeroMQ), & live YouTube Pub-Sub Model (RabbitMQ). Deployed & Analysed on Google Cloud using dockers, with 10 live clients

Enhancing Hallucination Detection | Natural Language Processing, LLMs

Feb,24 - Apr,24

- Innovated novel architecture from scratch for effective hallucination detection in Natural Language Generation tasks. Enhanced qualitative scores from **0.5 to 0.7** through cross-attention mechanisms & convolution layers.
- Integrated MetaModeling techniques using XGBoost, with 8 fine-tuned pretrained **LLMs** (7B & 11B params) & Reinforced Learning to **boost scores to 83%** in detecting hallucinations, improving semantic coherence.

QUIZZLE - Context-Based Quiz Generator | AI, ML, DL, NLP

Mar,23 - Apr,23

- Developed a software to design a quiz with variety of complete questions & answers, on any subject & topic.
- Used complex NLP & AI (LLMs) Models like **T5 Transformers, BARD, OpenAI, LLAMA**. Utilised **4+ GB** of Sense2Vec vectors through complex **Graph algorithms** for understanding context and generation.

CERTIFICATIONS

- Cloud Computing - Google Cloud
- GANs: Generative AI- Coursera
- Data Science - Jovian
- Full Stack Web Development - Udemy
- Applied ML: MLOPs - Coursera
- Internet of Things - NPTEL

ACHIEVEMENTS

- **RIISE-23 Conference** (Research ShowCase) for our work on WorkStealing at HiPEC Labs.